3.10 Apply nodal analysis to find node voltages V_1 to V_3 in the circuit of **Fig. P3.10** and then determine I_x.

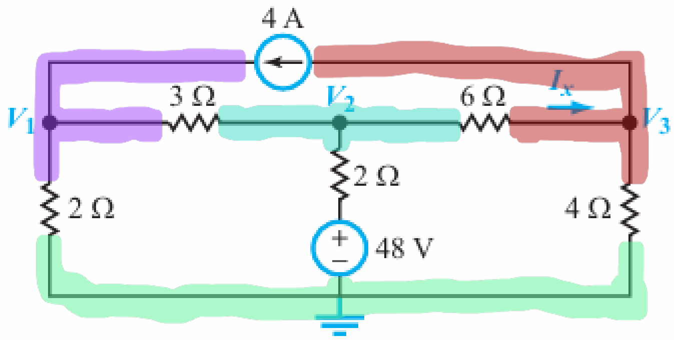


Figure P3.10 Circuit for Problem 3.10.

$$\text{KCL1:} \left[\frac{V_1}{2[\Omega]} + \frac{V_1 - V_2}{3[\Omega]} - 4[A]\right] = 0 \cdot 6[\Omega]$$

$$\Rightarrow 3V_1 + 2V_1 - 2V_2 - 24[V] = 0$$

$$= 5V_1 - 2V_2 = 24[V]$$

$$\text{KCL2:} \left[\frac{V_2 - V_1}{3[\Omega]} + \frac{V_2 - 48[V]}{2[\Omega]} + \frac{V_2 - V_3}{6[\Omega]}\right] = 0 \cdot 6[\Omega]$$

$$= 2V_2 - 2V_1 + 3V_2 - 144[V] + V_2 - V_3 = 0$$

$$\Rightarrow 6V_2 - 2V_1 - V_3 = 144[V]$$

$$\text{KCL 3:} \left[4[A] + \frac{V_3 - V_2}{6[\Omega]} + \frac{V_3}{4[\Omega]}\right] = 0 \cdot 12[\Omega]$$

$$= 48[V] + 2V_3 - 2V_2 + 3V_3 = 0$$

$$= 5V_3 - 2V_2 = -48\,[V]$$

$$\underbrace{\begin{bmatrix} 5 & -2 & 0 \\ -2 & 6 & -1 \\ 0 & -2 & 5 \end{bmatrix}}_{\text{Matrix A}} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} \underbrace{\begin{bmatrix} +24 \\ 144 \\ -48 \end{bmatrix}}_{\text{Matrix B}} = \boxed{\begin{bmatrix} 16.8 \\ 30 \\ 2.4 \end{bmatrix} \begin{array}{ll} \frac{84}{5} & V_1 \\ 30 & V_2 \\ \frac{12}{5} & V_3 \end{array}}$$

$$V_1 = 16.8\,[V],\ V_2 = 30\,[V],\ V_3 = 2.4\,[V]$$

$$I_x = \frac{V_2 - V_3}{6\,[\Omega]} = \frac{30 - 2.4}{6\,[\Omega]} = \boxed{4.6\,[A]}$$

3.12 The magnitude of the dependent current source in the circuit of **Fig. P3.12** depends on the current I_x flowing through the 10 Ω resistor. Determine I_x.

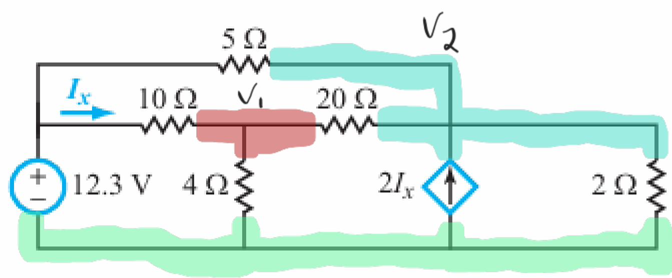


Figure P3.12 Circuit for Problems 3.12 and 3.13.

$$KCL1: \left[\frac{V_1 - 12.3[V]}{10[\Omega]} + \frac{V_1}{4[\Omega]} + \frac{V_1 - V_2}{20[\Omega]}\right] = 0 \cdot 20[\Omega]$$

$$\Rightarrow 2V_1 - 24.6[V] + 5V_1 + V_1 - V_2 = 0$$

$$= 8V_1 - V_2 = 24.6[V]$$

$$KCL2: \left[-2I_x + \frac{V_2}{2[\Omega]} + \frac{V_2 - 12.3}{5[\Omega]} + \frac{V_2 - V_1}{20[\Omega]}\right] = 0 \cdot 20\Omega$$

$$-2I_x = -2\left[\frac{12.3[V] - V_1}{10}\right] = \left[\frac{V_1 - 12.3}{5}\right] 20 = 4V_1 - 49.2$$

$$= 4V_1 - 49.2[V] + 10V_2 + 4V_2 - 49.2 + V_2 - V_1 = 0$$

$$\Rightarrow 15V_2 + 3V_1 = 98.4$$

$$\begin{bmatrix} 8 & -1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 24.6 \\ 32.8 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \Rightarrow \begin{matrix} 3.8\ [V] \\ 5.8\ [V] \end{matrix}$$

MatA MatB

$$I_x = \frac{12.3\,[V] - 3.8\,[V]}{10\,[\Omega]} = \boxed{0.85\,[A]}$$

3.33 Apply mesh analysis to the circuit in **Fig. P3.33** to determine V_x.

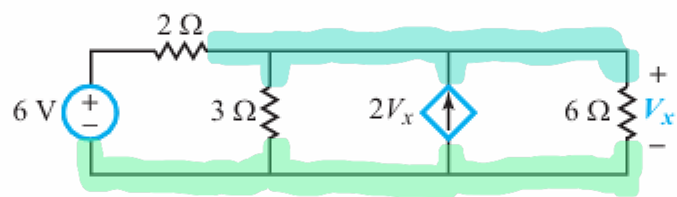


Figure P3.33 Circuit for Problem 3.33.

$$KCL1: \left[\frac{V_1 - 6[V]}{2[\Omega]} + \frac{V_1}{3[\Omega]} - 2V_x + \frac{V_1}{6[\Omega]} \right] = 0 \quad 6[\Omega]$$

$$\Rightarrow 3V_1 - 18[V] + 2V_1 - 12V_1 + V_1 = 0$$

$$\Rightarrow 6V_1 - 12V_1 = 18[V]$$

$$\Rightarrow -6V_1 = 18[V] \Rightarrow -V_1 = 3[V] \Rightarrow \boxed{V_1 = -3[V]}$$

3.34 Use the supermesh concept to solve for I_x in the circuit of **Fig. P3.34**.

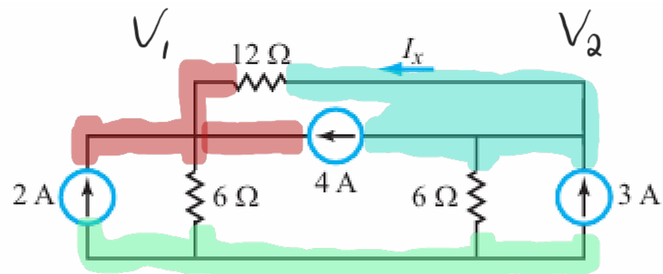


Figure P3.34 Circuit for Problem 3.34.

$$\text{KCL1: } \left[-2[A] + \frac{V_1}{6[\Omega]} - 4[A] + \frac{V_1 - V_2}{12[\Omega]}\right] = 0 \cdot 12\Omega$$

$$= -24[V] + 2V_1 - 48[V] + V_1 - V_2 = 0$$

$$\Rightarrow 3V_1 - V_2 = 72[V]$$

$$\text{KCL2: } \left[-3[A] + 4[A] + \frac{V_2}{6[\Omega]} + \frac{V_2 - V_1}{12[\Omega]}\right] = 0 \; 12[\Omega]$$

$$= -36[V] + 48[V] + 2V_2 + V_2 - V_1 = 0$$

$$\Rightarrow 3V_2 - V_1 = -12[V]$$

$$\underbrace{\begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}}_{\text{MatA}} \quad \underbrace{\begin{bmatrix} 72 \\ -12 \end{bmatrix}}_{\text{MatB}} = \begin{array}{c|c} \begin{bmatrix} 25.5 \\ 4.5 \end{bmatrix} & \begin{matrix} V_1 \\ V_2 \end{matrix} \end{array}$$

$$\boxed{I_x = \frac{V_2 - V_1}{12\Omega} = \frac{4.5 - 25.5}{12} = -1.75[A]}$$

Repeated Problem, Already Solved

3.37 The circuit in **Fig. P3.37** includes a dependent current source. Apply mesh analysis to determine I_x.

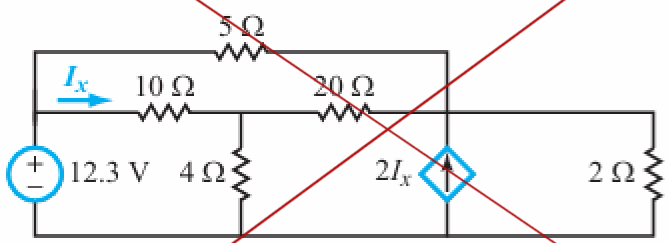


Figure P3.37 Circuit for Problems 3.37 and 3.38.

3.71 Find the Thévenin equivalent circuit at terminals (a, b) of the circuit in **Fig. P3.71**.

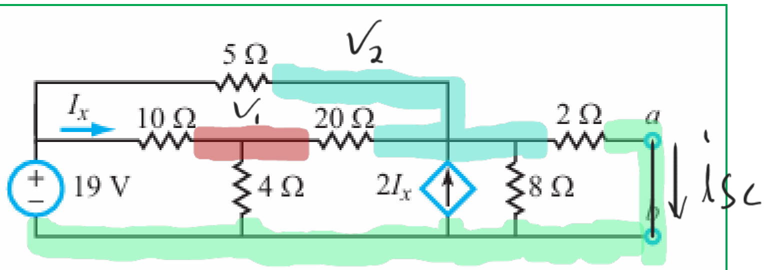


Figure P3.71 Circuit for Problems 3.71 and 3.72.

$$KCL: \left[\frac{V_1 - 19[V]}{10[\Omega]} + \frac{V_1}{4[\Omega]} + \frac{V_1 - V_2}{20[\Omega]} \right] = 0 \cdot 20\Omega$$

$$= 2V_1 - 38[V] + 5V_1 + V_1 - V_2 = 0$$

$$= 8V_1 - V_2 = 38[V]$$

$$KCL2: \left[\frac{V_2 - V_1}{20[\Omega]} - 2I_x + \frac{V_2}{8[\Omega]} + \frac{V_2 - 19[V]}{5[\Omega]} \right] = 0 \cdot 40\Omega$$

$$= 2V_2 - 2V_1 - 40I_x + 5V_2 + 8V_2 - 152[V] = 0$$

$$\Rightarrow 15V_2 - 2V_1 - 40I_x = 152[V]$$

$$I_x = \frac{19 - V_1}{10} \cdot -40 = [4V_1 - 76[V]]2 = 8V_1 - 152$$

$$\Rightarrow 15V_2 + 6V_1 = 304[V]$$

$$\begin{bmatrix} 8 & -1 \\ 6 & 15 \end{bmatrix} \quad \begin{bmatrix} 38 \\ 304 \end{bmatrix} = \begin{array}{c|c} \begin{bmatrix} 6.934 \\ 17.492 \end{bmatrix} & \begin{array}{c} V_1 \\ V_2 \end{array} \end{array}$$

Mat A $\qquad$ Mat B

$$V_{ab} = V_2 = 17.492\,[V]$$

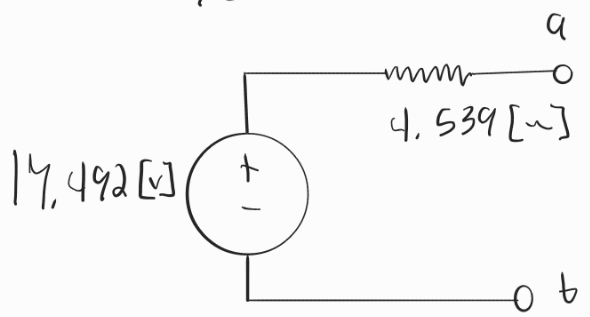
KCL For i_{sc}: ~~A~~ Just add $\frac{V_2}{2\Omega}$ to KCL2:

$$\left[\frac{V_2}{2\Omega}\right] \cdot 40\Omega = 20 V_2$$

$$\begin{bmatrix} 8 & -1 \\ 6 & 35 \end{bmatrix} \quad \begin{bmatrix} 38 \\ 304 \end{bmatrix} = \begin{array}{c|c} \begin{bmatrix} 5.413 \\ 4.406 \end{bmatrix} & \begin{array}{c} V_1 \\ V_2 \end{array} \end{array}$$

$$i_{sc} = \frac{4.406\,[V]}{2\,[\Omega]} = 3.853\,[A]$$

$$R_{Th} = \frac{17.492\,[V]}{3.853\,[A]} = 4.539\,[\Omega]$$



*3.74 Find the Norton equivalent circuit at terminals (a, b) of the circuit in **Fig. P3.74**.

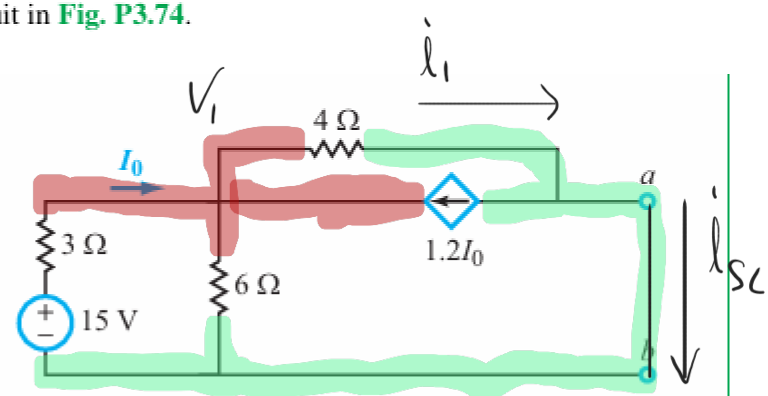


Figure P3.74 Circuit for Problems 3.74 and 3.75.

$$\text{KCL 1: } \left[\frac{V_1 - 15[V]}{3[\Omega]} + \frac{V_1}{6[\Omega]} + \frac{V_1}{4[\Omega]} - 1.2 I_0\right] = 0 \cdot 12[\Omega]$$

$$\Rightarrow 4V_1 - 60[V] + 2V_1 + 3V_1 - 14.4 I_0 = 0$$

$$\Rightarrow 9V_1 - 60[V] - 14.4 I_0 = 0$$

$$I_0 = \left[\frac{15[V] - V_1}{3[\Omega]}\right] \cdot 14.4 = 72[V] - 4.8V_1$$

$$= 13.8V_1 - 132[V] = 0 \Rightarrow V_1 = \frac{132[V]}{13.8} = 9.565[V]$$

$$i_1 = \frac{9.565[V]}{4[\Omega]} = 2.391[A] \qquad i_0 = \frac{15 - 9.565}{3} = 1.811[A] \cdot 1.2 = 2.174[A]$$

$$i_{sc} = i_1 - 1.2 i_0 = (2.391 - 2.174)[A] = \boxed{217[mA]}$$

*4.23 Find the value of v_o in the circuit in Fig. P4.23.

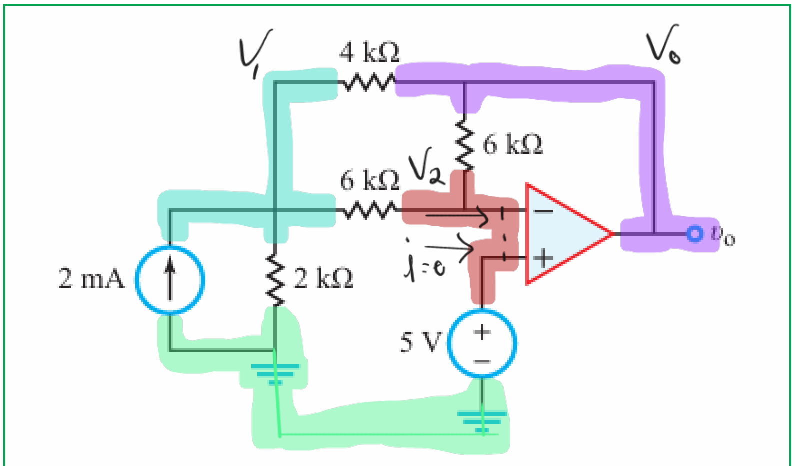


Figure P4.23 Circuit for Problem 4.23.

$$\text{KCL } V_2: \left[\frac{V_2 - V_1}{6\,[k\Omega]} + \frac{V_2 - V_0}{6\,[k\Omega]}\right] = 0 \cdot 6\,[k\Omega]$$

$$= 2V_2 - V_1 - V_0 = 0,\ V_2 = 5\,[V] = 10\,[V] - V_1 - V_0 = 0$$

$$\Rightarrow V_0 = 10\,[V] - V_1$$

$$\text{KCL } V_1: \left[-2\,[mA] + \frac{V_1}{2\,[k\Omega]} + \frac{V_1 - V_2}{6\,[k\Omega]} + \frac{V_1 - V_0}{4\,[k\Omega]}\right] = 0 \cdot 12\,[k\Omega]$$

$$= -24\,[V] + 6V_1 + 2V_1 - 2V_2 + 3V_1 - 3V_0 = 0$$

$$= 11V_1 - 2V_2 - 3V_0 - 24\,[V] = 0 \quad \text{Plug in } V_2 = 5\,[V],\ V_0 = 10\,[V] - V_1$$

$$11V_1 - 3V_0 - 24\,[V] - 3\left[10\,[V] - V_1\right] - 2\left(5\,[V]\right)$$

$$\Rightarrow 14V_1 - 64\,[V] = 0\,[V],\ V_1 = \frac{64\,[V]}{14} = 4.571\,[V]$$

$$V_0 = 10[V] - V_1 \Rightarrow 10[V] - 4.571[V] = 5.429[V]$$

$$\boxed{V_0 = 5.429\ [V]}$$

4.24 For the circuit in **Fig. P4.24**, obtain an expression for voltage gain $G = \upsilon_o / \upsilon_s$.

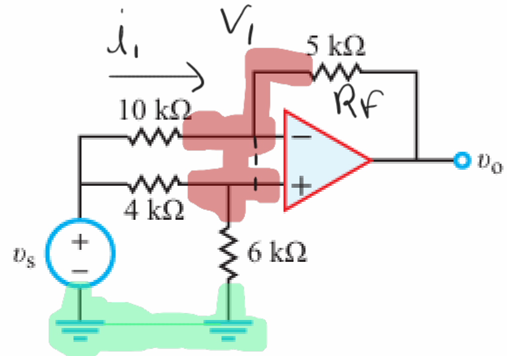


Figure P4.24 Circuit for Problem 4.24.

$$V_1 = V_s\left(\frac{6\,k\Omega}{(6+4)\,k\Omega}\right) \Rightarrow V_1 = 0.6\,V_s$$

$$\text{KCL: }\left[\frac{0.6V_s - V_s}{10\,[k\Omega]} + \frac{0.6V_s - V_o}{5\,[k\Omega]}\right] = 0 \cdot 10\,[k\Omega]$$

$$= 0.6V_s - V_s + 1.2V_s - 2V_o = 0$$

$$= 0.8V_s - 2V_o = 0 \quad = V_o = 0.4V_s$$

$$V_{Rf} = V_o - V_1 = 0.4V_s - 0.6V_s = -0.2V_s$$

$$G = \frac{V_o}{V_s} = \frac{0.4V_s}{V_s} = \boxed{0.4}$$